

Differential: M1 supplement

PRV 201 Frontiers of Technology (AI) · Track A: Build · Stevens Institute of Technology · September 2026

This supplement accompanies the M1 problem statement. The statement stands on its own; nothing here is needed to read it.

Industry watch, September 2026: breakthroughs, launches, regulation and investment

- **Frontier models score highly on textbook analog questions.** The top Razavi-Bench model scored 95.50%, graded by another language model, and the benchmark has an experimental simulator-assisted mode [1]. A bench tool must add what textbooks do not test: tolerances, ambiguity between faults, the cost of each measurement and safety. T5 compares the full system with such a model working alone.
- **IBM Maximo 9.2 (June 2026)** adds agentic workflows and an MCP server through which customers can connect their own agents [2], a possible route for a diagnosis engine into industrial repair depots.
- **EU AI Act transparency rules apply from 2 August 2026** [3]; the 2026 amendments moved the Annex III high-risk rules to 2 December 2027 [4] and keep a duty on providers and deployers to support their staff's AI literacy [5]. They would bind Differential only once offered in the EU.
- **Colorado re-enacted its AI law** as a regime for automated decision-making technology in consequential decisions, approved on 14 May 2026 and in force from 1 January 2027 [6]. A shop that used repair tickets to decide about its staff could come within such rules; Differential's terms of use forbid that use.
- **Investment is flowing to AI for the trades, starting with dispatch.** Probook announced \$40 million on 23 June 2026, a \$34 million Series A led by Andreessen Horowitz and a \$6 million seed round led by Sequoia, for an AI operating system for HVAC, plumbing and electrical contractors that it built dispatch-first [7]. Our reading: investors are funding AI for scheduling and dispatch in repair businesses, while the diagnosis a technician does at the bench is not what this money buys; that is the gap Differential addresses, and repair businesses may meet AI first in their front office.

Course coverage map

Table 1. Course topics and where the project uses them, with parts named by title.

Wk	Topic	Where the project uses it	Depth
2	Introduction to AI/ML algorithms	Bayesian inference, Gaussian mixtures, boosted trees, information-gain measurement choice (Part 4 – Preliminary AI approach, risks and ethics)	Core
3	Python, the AI/ML lingua franca	The simulation, engine and evaluation code that produces every number in the statement (repository); no code appears in the statement itself	Core (code)
4	Introduction to large language models	Chat-model benchmark scores (Part 1 – Problem, stakeholders and industry context); the language-model baselines (Part 3 – Objectives and measurable success criteria; T5); why language models guess (Part 4 – Preliminary AI approach, risks and ethics)	Core
5	Building LLM-enhanced agents	An assistant whose tools act only on the diagnosis session; complaint reading and explanation (Part 4 – Preliminary AI approach, risks and ethics)	Supporting
6	Building ML systems	Pilots, locked targets with confidence bounds, stress sets, hardware protocol (Part 3 – Objectives and measurable success criteria)	Core
7	Introduction to natural language processing	Complaint reading with noise; sealed paraphrase bank (Part 3 – Objectives and measurable success criteria; T18)	Supporting
8	Image recognition systems	Meter-photo reading scored on value, unit and mode (T19)	Supporting
9	Facial recognition systems (accuracy, bias, deployment)	Error breakdown, as in face-recognition audits, by who bears the errors: complaint writers, technicians, owners of undocumented gear (Part 4 – Preliminary AI approach, risks and ethics)	Supporting
10	Autonomous transportation	Five levels of diagnostic autonomy, L0 (the technician does everything) to L4 (an unattended fixture), defined by analogy to driving automation; Differential placed at L2 on safety and accountability grounds (Part 4 – Preliminary AI approach, risks and ethics)	Supporting
11	Introduction to robotics	A sense-decide loop without actuation: instruments sense (SCPI on the 24 V boards), the engine decides the next measurement and a person acts; automated test fixtures (levels L3-L4) would add the actuation (Part 4 – Preliminary AI approach, risks and ethics)	Partial
12	AI in cybersecurity and fraud detection	Anomaly detection: the calibrated “no single fault fits” outcome flags units outside the catalog (Part 3 – Objectives and measurable success criteria; T13, T22); prompt injection, untrusted data, session-only tools and adversarial prompts (Part 4 – Preliminary AI approach, risks and ethics; T16)	Core
13	Ethics of AI: risks, responsibility and governance	Hazard audit, OSHA, liability, EU AI Act and product liability, research ethics, NIST AI RMF (Part 1 – Problem, stakeholders and industry context, Part 4 – Preliminary AI approach, risks and ethics)	Core

Secondary targets

Table 2. Secondary targets: same rule, reported, not deciding success. Micro-F1: symptom reading scored over all (complaint, symptom) pairs; AUROC: chance an out-of-catalog unit gets a higher “no single fault fits” probability than a single-fault unit.

ID	Secondary target	Bar	ID	Secondary target	Bar
T2	Named first, channel strip only	≥ 94%	T24	Powered readings above 50 V per diagnosis ÷ half-split’s	≤ 0.19×
T3	True group among the first three shown	≥ 99%	T14	Accuracy change with a wrong prior, either way	≥ −1/−1 points
T8	Confirmed-answer effort ÷ fixed-order chart’s	≤ 0.64×	T18	Symptom-reading micro-F1 on a sealed bank (API key)	≥ 0.90 (Claude); ≥ 0.80 (rules)
T10	Confirmed-answer effort ÷ random probing’s	≤ 0.59×	T19	Exact reading of ≥ 200 meter photos (API key)	≥ 95% (Claude); ≥ 90% (offline)
T11	Confirmed-answer effort, complaint read ÷ ignored	≤ 0.98×	T20	95th-percentile time per recommendation	≤ 0.5 s
T22	Out-of-model units flagged; separation (AUROC)	≥ 51% flagged; AUROC ≥ 0.77	T21	Mean API cost per diagnosis at list prices (API key)	≤ \$0.10
T23	Single faults named; accuracy when naming	≥ 94% named; ≥ 99% right when naming			

How each bar was set

Every bar that a pilot can estimate follows one rule, fixed before the pilot ran: the pilot estimate moved 1.5 standard errors against the system, rounded to the reporting step. One refinement came after the pilot and before any test unit: rounding never carries a bar past the pilot estimate, and effort ratios round to 0.01 instead of 0.05. The coarser step had rounded T9’s bar onto its own pilot estimate, and nearest rounding had set T23’s accuracy bar at 100%, which no confidence bound

can clear; the refinement loosened those two and tightened T8, T10 and T11. A target is met only when the test's one-sided 95% confidence bound clears the bar. The chance column counts both the pilot's own error and the test's. T4, T14, T22 and T23 have two or three parts; their derivations are in `results/targets.json`. T5 (no pilot without an API key) and T7 (hardware) were fixed in advance; T15-T17 are release gates.

Table 3. Bar derivations from the pilot (weighted like the test set). SE: standard error (points for rates). Raw bar: estimate moved 1.5 SE against the system; the bar rounds it to the reporting step, never past the estimate. For rates, “a pass needs” is the count whose one-sided bound clears the bar. The chance that every pilot-estimable primary is met, if they were independent, is about 25%; about 1 are expected to miss.

ID	Target	Pilot	SE	Raw bar	Bar	A pass needs	Chance
T1	Failed part's group named first	95.6%	1.0	94.0%	$\geq 94\%$	at least 4728 of 5,000	83%
T2	Named first, channel strip only	96.5%	1.4	94.3%	$\geq 94\%$	at least 2370 of 2,500	88%
T3	True group among the first three shown	99.7%	0.6	98.7%	$\geq 99\%$	at least 4962 of 5,000	76%
T6	Failed part's group named first, aged units	93.2%	1.2	91.3%	$\geq 91\%$	at least 4584 of 5,000	88%
T13	Out-of-catalog units sent only to healthy parts	8.8%	2.0	11.8%	$\leq 12\%$	at most 272 of 2,500	84%
T9	Confirmed-answer effort $\div$ half-split tracing's	0.651	0.012	0.669	$\leq 0.67\times$	bound clears the bar	85%
T8	Confirmed-answer effort $\div$ fixed-order chart's	0.625	0.012	0.643	$\leq 0.64\times$	bound clears the bar	76%
T10	Confirmed-answer effort $\div$ random probing's	0.572	0.011	0.588	$\leq 0.59\times$	bound clears the bar	86%
T11	Confirmed-answer effort, complaint read $\div$ ignored	0.965	0.010	0.980	$\leq 0.98\times$	bound clears the bar	84%
T12	Calibration error of the shown probabilities at the stop	0.012	0.006	0.020	≤ 0.02	bound clears the bar	81%

References

- [1] Z. Zhang, “Razavi-Bench: An Expert-Curated Benchmark for Analog-Design Reasoning,” *Benchmark repository and leaderboard (GitHub Arcadia-1/razavi-bench)*, README at commit c8bce43, September 29 snapshot, Sep. 2026. [Online]. Available: <https://github.com/Arcadia-1/razavi-bench> (accessed Sep. 29, 2026).
- [2] IBM, “Introducing Maximo Application Suite 9.2: AI built for asset management that keeps operations running,” Press release, Jun. 25, 2026. [Online]. Available: <https://www.ibm.com/new/announcements/introducing-maximo-application-suite-9-2> (accessed Sep. 28, 2026).
- [3] European Commission, “Commission starts enforcing AI Act rules and new transparency requirements on 2 August (IP/26/1714),” Press release, Jul. 31, 2026. [Online]. Available: <https://digital-strategy.ec.europa.eu/en/news/commission-starts-enforcing-ai-act-rules-and-new-transparency-requirements-2-august> (accessed Sep. 28, 2026).
- [4] European Parliament and Council of the European Union, “Regulation (EU) 2026/1744 of 8 July 2026 amending Regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as regards the simplification of the implementation of harmonised rules on artificial intelligence (Digital Omnibus on AI),” *Official Journal of the European Union, OJ L*, 2026/1744, Jul. 24, 2026. [Online]. Available: <https://eur-lex.europa.eu/eli/reg/2026/1744/oj/eng>.
- [5] Publications Office of the European Union, “Regulation (EU) 2024/1689, consolidated text 02024R1689-20260727 (incorporating Regulation (EU) 2026/1744),” *EUR-Lex, consolidated text 02024R1689-20260727*, Jul. 27, 2026. [Online]. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:02024R1689-20260727>.
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